

**DANSVILLE, NY, USA**WMO: **724988**Lat: **42.571N**Lon: **77.713W**Elev: **209**StdP: **98.84**Time Zone: **-5.00 (NAE)**Period: **01-19**WBAN: **94704****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.6      | -13.9 | -21.5                         | 0.6 | -15.0 | -18.9 | 0.7 | -13.0 | 9.8                   | -2.8 | 8.8 | -3.0 | 2.7                   | 160  | 0.501 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.6                   | 32.4            | 22.4 | 31.0 | 21.6 | 29.4 | 21.0 | 23.9                | 29.7 | 23.0 | 28.5 | 22.3 | 27.4 | 3.2                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.2                            | 17.3 | 26.6 | 21.3 | 16.3 | 25.8 | 20.6 | 15.6 | 25.2 | 72.1          | 29.8 | 69.0 | 28.5 | 66.2 | 27.5 | 26.8              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |            |            | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |              |             |              |             |              |             |    |    |
|-------------------|------------|------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|--------------|-------------|--------------|-------------|--------------|-------------|----|----|
|                   |            |            | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years   |             | n=20 years   |             | n=50 years   |             |    |    |
| 1%                | 2.5%       | 5%         | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min          | Max         | Min          | Max         | Min          | Max         | DB | WB |
| (a)               | (b)        | (c)        | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)          | (l)         | (m)          | (n)         | (o)          | (p)         |    |    |
| (4) <b>8.5</b>    | <b>7.5</b> | <b>6.7</b> | <b>-20.0</b>               | <b>34.4</b> | <b>2.5</b>         | <b>1.6</b> | <b>-21.8</b>                                       | <b>35.6</b> | <b>-23.3</b> | <b>36.5</b> | <b>-24.7</b> | <b>37.4</b> | <b>-26.5</b> | <b>38.6</b> |    |    |
| (5)               |            |            | <b>-20.4</b>               | <b>25.2</b> | <b>2.4</b>         | <b>1.1</b> | <b>-22.1</b>                                       | <b>26.0</b> | <b>-23.5</b> | <b>26.7</b> | <b>-24.8</b> | <b>27.3</b> | <b>-26.6</b> | <b>28.1</b> |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.1   | -3.2  | -1.8  | 2.1   | 8.8   | 15.3  | 19.9  | 22.4  | 21.3  | 17.9  | 11.5  | 5.5   | 0.5  |
|                                                                                  | DBStd       | 10.22  | 6.67  | 5.67  | 6.13  | 5.43  | 5.05  | 3.70  | 2.98  | 2.96  | 3.92  | 4.81  | 5.06  | 5.14 |
|                                                                                  | HDD10.0     | 1586   | 412   | 333   | 257   | 87    | 9     | 0     | 0     | 0     | 0     | 39    | 152   | 297  |
|                                                                                  | HDD18.3     | 3398   | 668   | 565   | 504   | 291   | 123   | 26    | 3     | 8     | 54    | 217   | 386   | 554  |
|                                                                                  | CDD10.0     | 1613   | 2     | 2     | 11    | 49    | 174   | 296   | 386   | 351   | 238   | 87    | 16    | 2    |
|                                                                                  | CDD18.3     | 382    | 0     | 0     | 0     | 3     | 29    | 72    | 130   | 101   | 42    | 6     | 0     | 0    |
|                                                                                  | CDH23.3     | 3507   | 0     | 0     | 4     | 63    | 317   | 688   | 1210  | 798   | 370   | 56    | 1     | 0    |
|                                                                                  | CDH26.7     | 1148   | 0     | 0     | 0     | 15    | 99    | 232   | 434   | 246   | 113   | 9     | 0     | 0    |
| Wind                                                                             | WSAvg       | 2.9    | 3.2   | 3.2   | 3.4   | 3.4   | 2.9   | 2.7   | 2.4   | 2.4   | 2.5   | 2.8   | 2.9   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 731    | 37    | 32    | 42    | 61    | 68    | 85    | 77    | 79    | 81    | 62    | 64    | 47   |
|                                                                                  | PrecMax     | 987    | 94    | 79    | 78    | 118   | 178   | 251   | 190   | 169   | 180   | 143   | 140   | 98   |
|                                                                                  | PrecMin     | 525    | 5     | 2     | 15    | 23    | 18    | 10    | 13    | 27    | 12    | 4     | 11    | 7    |
|                                                                                  | PrecStd     | 128    | 21    | 16    | 17    | 23    | 35    | 43    | 38    | 34    | 34    | 31    | 29    | 21   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.1  | 17.2  | 22.9  | 28.8  | 31.8  | 33.2  | 34.2  | 33.3  | 32.5  | 27.8  | 21.2  | 17.1 |
|                                                                                  |             | MCWB   | 12.0  | 10.5  | 14.9  | 17.3  | 20.6  | 22.6  | 23.6  | 22.9  | 22.2  | 19.6  | 14.0  | 12.1 |
|                                                                                  | 2%          | DB     | 12.4  | 12.5  | 18.0  | 25.0  | 29.6  | 31.4  | 32.4  | 31.2  | 29.8  | 24.8  | 18.0  | 12.8 |
|                                                                                  |             | MCWB   | 9.4   | 7.8   | 11.6  | 15.4  | 19.3  | 21.6  | 22.9  | 22.1  | 21.4  | 17.2  | 12.6  | 9.7  |
|                                                                                  | 5%          | DB     | 8.7   | 9.2   | 14.6  | 21.6  | 27.0  | 29.5  | 30.9  | 29.4  | 27.5  | 22.1  | 16.2  | 10.1 |
|                                                                                  |             | MCWB   | 6.1   | 5.5   | 9.7   | 13.3  | 18.6  | 20.9  | 22.0  | 21.4  | 20.1  | 16.9  | 11.6  | 7.3  |
|                                                                                  | 10%         | DB     | 6.0   | 6.0   | 11.8  | 18.6  | 24.4  | 27.4  | 29.1  | 27.7  | 25.1  | 19.2  | 13.5  | 7.7  |
|                                                                                  |             | MCWB   | 3.6   | 2.8   | 7.4   | 12.1  | 16.9  | 19.8  | 21.2  | 20.8  | 19.0  | 14.9  | 9.9   | 5.3  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 12.6  | 12.3  | 15.8  | 18.8  | 22.5  | 24.0  | 25.2  | 24.5  | 24.0  | 21.1  | 15.7  | 13.4 |
|                                                                                  |             | MCDB   | 15.4  | 15.7  | 21.3  | 26.0  | 29.7  | 30.4  | 31.9  | 30.4  | 29.5  | 25.3  | 18.6  | 16.2 |
|                                                                                  | 2%          | WB     | 9.3   | 8.5   | 12.9  | 16.4  | 20.8  | 22.8  | 24.0  | 23.5  | 22.4  | 19.1  | 13.7  | 10.1 |
|                                                                                  |             | MCDB   | 11.7  | 12.0  | 16.7  | 22.9  | 27.1  | 29.1  | 30.1  | 28.7  | 27.7  | 22.9  | 17.4  | 12.6 |
|                                                                                  | 5%          | WB     | 6.2   | 5.7   | 10.4  | 14.7  | 19.7  | 21.9  | 23.2  | 22.6  | 21.2  | 17.1  | 12.1  | 7.6  |
|                                                                                  |             | MCDB   | 8.6   | 9.2   | 13.8  | 20.6  | 25.0  | 27.8  | 28.6  | 27.5  | 25.8  | 21.0  | 15.4  | 9.7  |
|                                                                                  | 10%         | WB     | 3.6   | 2.9   | 7.7   | 12.6  | 18.3  | 20.8  | 22.3  | 21.7  | 20.0  | 15.4  | 10.2  | 5.5  |
|                                                                                  |             | MCDB   | 5.9   | 5.5   | 11.6  | 17.5  | 23.1  | 25.6  | 27.3  | 26.2  | 24.1  | 18.6  | 13.3  | 7.7  |
| Mean Daily<br>Temperature<br>Range                                               |             | MDBR   | 8.4   | 9.0   | 10.2  | 12.0  | 12.5  | 11.6  | 11.6  | 11.2  | 11.6  | 10.1  | 9.2   | 7.3  |
|                                                                                  | 5% DB       | MCDBR  | 11.0  | 13.5  | 14.5  | 17.4  | 15.0  | 14.0  | 13.3  | 13.1  | 13.9  | 13.9  | 13.4  | 11.0 |
|                                                                                  |             | MCWBR  | 8.8   | 9.9   | 9.3   | 9.9   | 7.2   | 6.5   | 5.7   | 5.8   | 6.7   | 7.9   | 8.9   | 8.5  |
|                                                                                  | 5% WB       | MCDBR  | 10.5  | 12.6  | 13.1  | 15.2  | 13.1  | 12.0  | 11.3  | 10.9  | 11.7  | 12.1  | 11.5  | 10.3 |
|                                                                                  |             | MCWBR  | 8.8   | 10.0  | 9.0   | 9.4   | 6.9   | 6.1   | 5.4   | 5.4   | 6.1   | 7.8   | 8.7   | 8.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.308  | 0.323 | 0.347 | 0.379 | 0.402 | 0.448 | 0.446 | 0.436 | 0.376 | 0.359 | 0.315 | 0.297 |      |
|                                                                                  | taud        | 2.349  | 2.307 | 2.338 | 2.307 | 2.286 | 2.182 | 2.196 | 2.214 | 2.368 | 2.409 | 2.499 | 2.455 |      |
|                                                                                  | Ebn at Noon | 825    | 872   | 894   | 888   | 873   | 830   | 827   | 823   | 853   | 822   | 820   | 813   |      |
|                                                                                  | Edh at Noon | 82     | 101   | 111   | 124   | 130   | 145   | 142   | 135   | 107   | 91    | 71    | 69    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.34   | 2.17  | 3.29  | 4.41  | 5.29  | 5.63  | 5.71  | 5.13  | 4.12  | 2.53  | 1.60  | 1.09  |      |
|                                                                                  | RadStd      | 0.12   | 0.26  | 0.34  | 0.48  | 0.54  | 0.48  | 0.46  | 0.30  | 0.40  | 0.26  | 0.22  | 0.16  |      |